Enhancing Inclusive Education in Sri Lanka Through a Personalized Multimodal Learning Platform for Children with Neurodiversity

ID: 25-26J-218

Project Proposal Report

Author: Wijesuriya M W A D

(IT22369574)

B.Sc. (Hons) Degree in Information Technology, Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

AI-Powered Predictive Progress Tracking and Feedback System for Students with Down Syndrome

Project ID: 25-26J-218

Project Proposal Report

Author: Wijesuriya M W A D

(IT22369574)

Supervisor: Prof. Hansika Mahaadikaram

Co-Supervisor: Ms. Poorna Panduwawala

B.Sc. (Hons) Degree in Information Technology, Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text

Name	Student ID	Signature
M. W. A. D. Wijesuriya	IT22369574	

The above candidate is conducting research for the undergraduate Dissertation under my supervision.

.....

Signature of the supervisor

Ms. Hansika Mahaadikara

.....

Signature of the co-supervisor

Ms. Poorna Panduwawala

4/25/2026

.....

Date

4/25/2026

.....

Date

ABSTRACT

Children with Down syndrome face significant challenges in areas such as attention regulation, cognitive processing, and learning engagement, requiring continuous monitoring and timely intervention to support effective learning outcomes. However, most existing educational systems rely on retrospective progress reports and lack predictive capabilities to identify early signs of disengagement or learning fatigue. This research proposes an AI-powered predictive progress tracking and parental–teacher feedback system specifically designed for students with Down syndrome in Sri Lanka. The system captures real-time learner interaction data, including task completion, time spent, and behavioral engagement patterns, and analyzes these data using machine learning models to predict potential attention decline and learning difficulties. Unlike traditional monitoring tools, the proposed system generates intelligent early-warning alerts and presents simplified, accessible dashboards for parents and teachers, enabling proactive and informed intervention. The system architecture integrates a user-friendly front-end interface with a secure backend analytics engine, ensuring real-time processing, data privacy, and scalability. By transforming passive monitoring into a predictive and caregiver-centered support mechanism, this research aims to enhance learner engagement, improve intervention timing, and strengthen collaboration between parents and teachers, demonstrating the potential of AI-driven technologies to advance inclusive education.

Keywords: Down syndrome, AI, Predictive Analytics, Progress Tracking, Inclusive Education, Early Intervention

ACKNOWLEDGEMENT

The author would like to express sincere gratitude to Prof. Hansika Mahaadikaram (Supervisor) and Ms. Poorna Panduwawala (Co-Supervisor) for their invaluable guidance, continuous support, and constructive feedback throughout this research. Appreciation is also extended to the Sri Lanka Institute of Information Technology (SLIIT) and the Department of Information Technology for providing the necessary resources and academic environment. The author is thankful to the teachers, parents, and caregivers who contributed insights to this study. Finally, heartfelt thanks are given to family and friends for their encouragement and unwavering support.

Table of Contents

DECLARATION **2**

ABSTRACT **3**

ACKNOWLEDGEMENT **4**

LIST OF FIGURES **6**

LIST OF TABLES **7**

LIST OF ABBREVIATIONS **8**

1. INTRODUCTION **1**

1.1. Background & Literature Survey **2**

1.2. Research Gap **3**

1.3. Research Problem **7**

1.4. Research Objective **9**

1.4.1. Main Objectives **9**

1.4.2. Specific Objectives **9**

2. METHODOLOGY **10**

2.1. Methodology **10**

2.3. Overall System Description ..... Error! Bookmark not defined.

3. Testing **18**

4. Use Case Diagram **22**

5. PROJECT REQUIREMENTS **24**

5.1. Functional Requirements **24**

5.2. Non-Functional Requirements **24**

6. SOCIAL, ETHICAL, AND SECURITY CONSIDERATIONS **26**

Social Impact **26**

Ethical Considerations **26**

Security Considerations **27**

7. GANTT CHART **36**

8. WORK BREAKDOWN STRUCTURE **37**

9. BUDGET **38**

10. Commercialization **39**

Reference list **40**

LIST OF FIGURES

Figure 1: Explanation of methodology 8
Figure 2: Overall system description 9
Figure 3: Individual use case diagram 12
Figure 4: Gantt chart 14
Figure 5: Work breakdown structure 15

LIST OF TABLES

Table 1: List of abbreviations viii
Table 2: Existing platforms and gaps.....3
Table 3: Existing platforms and gaps (based on Sri Lanka) 4
Table 4: Budget..... 16

LIST OF ABBREVIATIONS

Table 1: List of abbreviations

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
AR	Augmented Reality
CNN	Convolutional Neural Network
GPS	Global Positioning System
ML	Machine Learning
UI/UX	User Interface / User Experience
AAC	Augmentative and Alternative Communication
AES	Advanced Encryption Standard
ANN	Artificial Neural Network
LKR	Sri Lankan Rupee
LMS	Learning Management System
PIN	Personal Identification Number
RBAC	Role-Based Access Control
SUS	System Usability Scale
TLS	Transport Layer Security
UDL	Universal Design for Learning
UNESCO	United Nations Educational, Scientific and Cultural Organization
AUC	Area Under the Curve

1. INTRODUCTION

Inclusive education emphasizes the importance of providing equitable and effective learning opportunities for all learners, including those with intellectual and developmental disabilities. Among these groups, children with Down syndrome often experience challenges related to cognitive processing, attention regulation, memory retention, and adaptive learning pace, which can significantly affect their educational outcomes if not adequately supported [1]. These challenges highlight the need for continuous monitoring and personalized learning approaches tailored to individual learning needs.

Recent advancements in Artificial Intelligence (AI) and machine learning have significantly transformed the field of educational technology by enabling adaptive and data-driven learning environments. AI-powered systems can analyze learner behavior, track engagement levels, and generate insights that support personalized instruction and early identification of learning difficulties [2], [3]. Such technologies have demonstrated strong potential in improving accessibility and learner engagement, particularly in inclusive education contexts [4].

However, most existing educational platforms and learning management systems primarily rely on retrospective performance evaluation, such as completed tasks and assessment scores. These systems fail to capture real-time behavioral indicators, including attention fluctuations, learning fatigue, and disengagement, which are critical factors affecting learners with Down syndrome [5]. Furthermore, the limited integration of predictive analytics and caregiver feedback mechanisms restricts the ability of parents and teachers to intervene proactively and effectively.

To address these limitations, this research proposes an AI-powered predictive progress tracking and parental feedback system specifically designed for students with Down syndrome. The proposed system leverages real-time learner interaction data and machine learning models to detect early signs of attention decline and learning challenges. By providing intelligent alerts and simplified dashboards for parents and teachers, the system supports timely intervention and informed decision-making. This approach aims to enhance learner engagement, improve educational outcomes, and strengthen collaboration between caregivers and educators in inclusive learning environments.

1.1. Background & Literature Survey

1.1.1 General Background

The integration of digital technologies into education has significantly transformed modern learning environments, particularly in the context of inclusive education. Advances in Artificial Intelligence (AI), machine learning, and educational data analytics have enabled the development of adaptive and personalized learning systems that respond to individual learner needs in real time [3], [6]. These technologies are increasingly recognized for their potential to support learners with intellectual and developmental disabilities by enhancing accessibility, engagement, and individualized instruction [7].

Children with Down syndrome and other neurodevelopmental conditions often exhibit unique learning characteristics, including reduced attention span, limited working memory, and slower cognitive processing speed. These challenges necessitate structured, repetitive, and closely monitored learning approaches to ensure effective knowledge acquisition [8]. Traditional teaching methods and conventional learning management systems are often insufficient, as they rely on periodic assessments and static performance reports, providing limited insight into real-time learner engagement and behavioral patterns.

Several existing educational platforms and technologies attempt to address aspects of learner engagement and monitoring. For instance, platforms such as ClassDojo [5] and DreamBox Learning [9] offer features for progress tracking, gamification, and parent–teacher communication. Gamification techniques, as discussed by Sebastian Deterding et al., have been shown to improve user motivation and engagement in learning environments [10]. Additionally, learning analytics frameworks provide insights into student performance and behavior, supporting data-driven decision-making in education [11].

Recent research has further explored the application of predictive analytics and AI in education. Predictive models can analyze learner interaction data to identify patterns related to disengagement, academic risk, and performance decline, enabling early intervention strategies [12], [13]. AI-driven dashboards and early warning systems have demonstrated effectiveness in identifying at-risk students and improving learning outcomes through timely alerts and actionable insights [14]. Furthermore, advancements in AI-based educational tools have enhanced parent–teacher communication and involvement, strengthening collaborative support for learners [15], [16].

Despite these advancements, existing systems exhibit several limitations. Most platforms primarily provide retrospective analysis rather than real-time predictive insights, limiting their ability to detect early signs of attention decline or learning fatigue. Additionally, many solutions are designed for general learner populations and lack customization for specific conditions such as Down syndrome. The absence of integrated systems that combine real-time behavioral monitoring, predictive analytics, and caregiver feedback remains a significant gap in current educational technologies.

Therefore, there is a clear need for an AI-powered, predictive, and caregiver-centered

learning support system tailored specifically for students with Down syndrome. Such a system should integrate real-time data collection, intelligent analytics, and user-friendly dashboards to enable proactive intervention and improve overall learning engagement in inclusive education environments.

1.1.2. Problem Context in Sri Lanka and Importance of AI in Education

Inclusive education for learners with neurodevelopmental conditions, such as Down syndrome, remains a significant challenge in developing countries like Sri Lanka. Although national policies promote equal educational opportunities, practical implementation is often limited due to the lack of specialized tools, trained educators, and structured support systems [22]. As a result, many learners with special needs struggle to receive personalized attention and continuous monitoring of their learning progress.

One of the major challenges in the Sri Lankan context is the limited availability of digital platforms specifically designed for learners with cognitive and developmental disabilities. Existing educational applications primarily focus on general learning and do not incorporate adaptive features tailored to individual learner needs. Furthermore, most systems lack real-time monitoring capabilities, making it difficult for caregivers to identify early signs of learning difficulties or reduced engagement.

Another key issue is the absence of caregiver-centered support mechanisms. In many cases, parents play a crucial role in supporting children with Down syndrome, especially in home-based learning environments. However, current systems provide minimal insights or guidance to caregivers, limiting their ability to make informed decisions regarding the child's learning progress [13]. This gap highlights the need for solutions that are not only learner-focused but also designed to assist caregivers through clear and actionable insights.

In addition, the shortage of domain-specific datasets poses a significant barrier to developing intelligent educational systems. Most machine learning models in education rely on large-scale datasets, which are often unavailable for special education contexts. This limitation reduces the effectiveness of traditional predictive systems and restricts their applicability in real-world scenarios.

Artificial Intelligence (AI) has emerged as a powerful tool to address these challenges by enabling data-driven decision-making and personalized learning experiences. AI techniques, particularly machine learning, allow systems to analyze learner behavior, identify patterns, and generate predictive insights that support early intervention [19]. Predictive analytics in education has been shown to improve learner outcomes by detecting performance issues before they become critical [3].

Moreover, AI-driven systems can adapt to individual learner needs by analyzing interaction data such as response time, accuracy, and engagement levels. This enables the development of personalized learning environments that cater to diverse cognitive abilities. Research has also highlighted the effectiveness of AI in supporting learners with neurodevelopmental disorders by providing adaptive interfaces and intelligent

feedback mechanisms [20].

Another important advantage of AI in education is its ability to enhance caregiver involvement. By transforming complex data into simple visual insights, AI systems can help parents better understand their child's learning behavior and progress. This aligns with recent studies emphasizing the importance of parental engagement in improving educational outcomes [13], [14].

In the Sri Lankan context, integrating AI into educational systems presents a promising opportunity to bridge the gap between limited resources and the growing need for inclusive education. By combining AI with game-based learning and real-time data collection, it is possible to create scalable and cost-effective solutions that support both learners and caregivers.

Therefore, this research leverages AI-driven predictive analytics and game-based data generation to address the identified challenges. The proposed approach aims to provide a practical, accessible, and adaptive learning support system tailored to the needs of learners with Down syndrome while empowering caregivers with meaningful insights.

1.2. Research Gap

Although the adoption of digital learning technologies and AI-driven educational systems has increased significantly, there remains a critical gap in solutions specifically designed to support learners with Down syndrome through parental-centered approaches. Most existing educational platforms primarily target general learner populations and institutional use, offering limited personalization for neurodiverse learners with unique cognitive and behavioral characteristics [6], [8].

Several widely used platforms, such as ClassDojo [5] and DreamBox Learning [6], provide features including progress tracking, gamification, and communication mechanisms. While these systems enhance engagement and visibility, they largely depend on retrospective performance data, such as completed tasks and assessment scores. Consequently, they lack the ability to capture real-time behavioral indicators such as attention fluctuation, learning fatigue, and disengagement, which are critical for supporting students with Down syndrome.

Recent advancements in predictive analytics demonstrate the potential of Artificial Intelligence (AI) to identify at-risk learners and forecast performance decline using behavioral data [10], [11]. Early warning systems and learning analytics dashboards have shown effectiveness in enabling data-driven interventions [13]. However, these systems are typically designed for institutional environments and do not provide simplified, accessible insights tailored for parents or guardians who support learners in home-based settings.

Furthermore, existing solutions often operate in a fragmented manner, where behavioral monitoring, predictive analytics, and communication features are implemented as separate components rather than as an integrated system. Although some platforms support communication between stakeholders [11], [15], they do not

offer a unified, parental-focused interface that combines real-time monitoring with predictive insights. This limitation reduces the ability of parents to understand learning patterns and respond proactively to potential issues.

In the Sri Lankan context, platforms such as Hapan, Nenasa, ATHWEL, and Avaz Sri Lanka provide various forms of educational and assistive support. However, these systems are largely domain-specific and lack integration, real-time behavioral tracking, and AI-driven predictive capabilities. Additionally, concerns related to data privacy, ethical AI usage, and parental trust remain significant barriers to adopting AI-based solutions in educational settings [17].

Therefore, there is a clear research gap in developing a unified, AI-powered system that integrates real-time behavioral data collection, predictive analytics, and simplified parental-focused feedback specifically tailored for students with Down syndrome. Addressing this gap will enable parents to better understand student learning behavior, support early intervention, and improve overall engagement in inclusive education environments.

Table 2: Comparison of Existing Systems and Identified Gaps

Feature	**Traditional LMS [9]**	**Adaptive Platforms (DreamBox) [12]**	**Communication Apps (ClassDojo) [16]**	**Proposed AI-Based System**
Real-time progress monitoring	✗	◐	✗	✓
Behavioral data tracking	✗	◐	✗	✓
AI-driven predictive analytics	✗	✗	✗	✓
Early alerts for attention/fatigue	✗	✗	✗	✓
Down syndrome–specific adaptation	✗	✗	✗	✓
Therapy-aligned monitoring	✗	✗	✗	✓
Dashboard interface	◐	◐	✓	✓
Parental feedback / support	✗	◐	✓	✓
Mobile-first implementation	◐	◐	✓	✓
Data security & role-based access	◐	◐	✓	✓
Suitability for Sri	◐	✗	✓	✓

Lanka context				

Table Legend

- ✓ : Fully supported
- ◐ : Partially supported
- ✗ : Not supported

Table 3: Comparison in Sri Lankan Context

Feature	Hapan	Nenasa	ATHWEL	Avaz Sri Lanka	Limitations in Current Sri Lankan Systems	Proposed System
Primary focus	General learning support	Educational TV / e-learning	Assistive learning	AAC communication	Systems are domain-specific, not integrated	Integrated learning + monitoring
Real-time progress monitoring	✗	✗	✗	✗	No continuous tracking of student behavior	✓
Behavioral data tracking (attention, engagement)	✗	✗	✗	◐	Limited behavioral insights	✓
AI-based predictive analytics	✗	✗	✗	✗	No predictive learning support	✓
Early alerts for intervention	✗	✗	✗	✗	No early warning mechanisms	✓
Down syndrome –specific support	✗	✗	◐	◐	No dedicated system	✓
Parental support / feedback	◐	✗	◐	◐	Limited structured parental insights	✓
Accessibility for special-needs learners	◐	◐	✓	✓	Partial, not personalized	✓
Mobile accessibility	✓	◐	✓	✓	Varies across platforms	✓

Ease of use	Moderate	Easy	Moderate	Easy	Not optimized for all users	Simple UI/UX
Affordability	✓	✓	✓	✓	Affordable but limited features	✓ (low-cost)
Integration of features	✗	✗	✗	✗	Fragmented solutions	✓ unified system

Table Legend

- ✓ : Fully supported
- ◐ : Partially supported
- ✗ : Not supported

1.3. Research Problem

The increasing adoption of digital learning platforms and learning analytics has improved the visibility of student performance; however, most existing systems remain inadequate for effectively supporting students with Down syndrome, who require continuous monitoring, structured guidance, and timely intervention. Current educational technologies primarily focus on general learner populations and rely on retrospective performance evaluation, such as completed tasks and assessment scores, rather than capturing real-time behavioral indicators that are critical for understanding learning engagement [8], [20].

Although platforms such as DreamBox Learning [6] and ClassDojo [5] provide features for adaptive learning and communication, they lack predictive capabilities to identify early signs of attention decline, learning fatigue, and disengagement. These limitations are particularly significant for students with Down syndrome, whose learning effectiveness depends on early detection of behavioral changes and timely support [10], [11].

Furthermore, recent advancements in Artificial Intelligence (AI) and predictive analytics demonstrate the potential to forecast learning difficulties and support proactive intervention strategies [3], [15]. However, most existing implementations are designed for institutional use and do not provide simplified, user-friendly insights tailored for parents or guardians. This creates a gap in enabling effective home-based support, where parental involvement plays a critical role in reinforcing learning outcomes.

In addition, existing systems often lack integration between behavioral monitoring, predictive analytics, and parental feedback mechanisms. While some platforms support communication between stakeholders [13], [14], they do not offer a unified system that combines real-time monitoring with intelligent alerts and actionable insights. This fragmentation reduces the ability of parents to understand their child's learning behavior and respond effectively to emerging challenges.

Moreover, in the Sri Lankan context, available educational and assistive platforms

such as Hapan, Nenasa, ATHWEL, and Avaz Sri Lanka provide limited functionality and are not equipped with AI-driven predictive features or real-time behavioral tracking. These systems are typically domain-specific and do not address the need for a comprehensive, accessible, and affordable solution tailored to children with Down syndrome.

Therefore, the central research problem addressed in this study is the lack of an AI-powered, predictive progress tracking and parental feedback system specifically designed for students with Down syndrome. There is a need for a unified, mobile-accessible solution that integrates real-time behavioral data collection, predictive analytics, and simplified parental insights to enable early intervention, improve learner engagement, and support effective inclusive education practices.

1.4. Research Objective

1.4.1. Main Objectives

The main objective of this research is to design and develop an AI-powered, predictive progress tracking and parental feedback system for students with Down syndrome, which enables real-time monitoring of learning behavior and provides simplified, data-driven insights to parents, supporting early intervention and improving overall learner engagement.

1.4.2. Specific Objectives

The specific objectives of this research are designed using the SMART (Specific, Measurable, Achievable, Realistic, Time-bound) framework to ensure practicality and successful evaluation during the final presentation and viva examination:

The proposed system will first focus on developing an AI-driven predictive tracking model capable of monitoring learner engagement, attention levels, and interaction patterns. The reliability and accuracy of these predictions will be evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score, and further validated through observational feedback from parents and domain experts [3], [20].

To ensure accessibility and inclusivity, the platform will incorporate simplified, neurodiversity-friendly user interfaces tailored to the needs of children with Down syndrome and their parents. The usability of these interfaces will be systematically evaluated using established methods such as the System Usability Scale (SUS) and task-completion rates to ensure ease of use and effectiveness [18], [19].

In addition, the system will integrate real-time behavioral monitoring mechanisms that continuously capture learner interaction data, including task completion, time spent, and engagement patterns. These data will be used to generate meaningful insights that help parents better understand their child's learning behavior and progress over time [8], [10].

The solution will also provide a parental-focused dashboard designed to deliver clear, actionable insights into learner performance while generating early warning alerts for potential attention decline or learning fatigue. The effectiveness of these features will be measured through parental engagement levels, response to alerts, and qualitative feedback collected during system usage [14], [18].

Finally, the overall system effectiveness will be validated through pilot testing within a controlled user group in Sri Lanka, with particular attention to improvements in learner engagement, attention consistency, and parental satisfaction. This evaluation will ensure that the system is practical, reliable, and suitable for real-world application in inclusive education environments [7], [8].

2. METHODOLOGY

2.1. Methodology

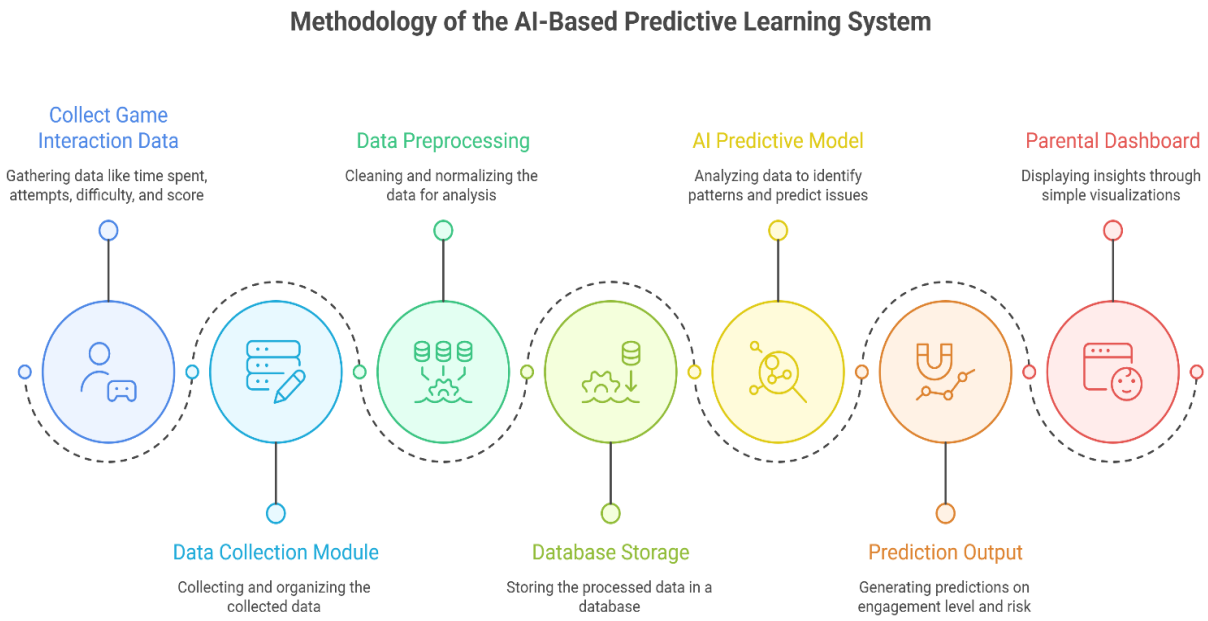

Figure 1: Explanation of methodology

The proposed system follows a data-driven methodology that integrates real-time data collection, predictive analytics, and parental feedback to support learners with Down syndrome. The overall process is designed to transform raw interaction data into meaningful insights that enable early intervention and improved learning engagement.

A game-based simulator was developed as part of the system to generate realistic learner interaction data. This simulator functions both as a learning environment and a data collection mechanism, ensuring that the dataset reflects real-world behavioral patterns. The generated data are used for training and evaluating the predictive model.

The overall process can be explained as follows:

Data Collection from Game Activities: Learner interaction data are collected while children engage with game-based activities within the application. The collected data include time spent on tasks, number of attempts, difficulty level of the game, and achieved scores. These data serve as the primary input for analyzing learner engagement and performance.

Data Processing and Storage: The collected data are stored in a centralized database. Preprocessing techniques such as data cleaning, normalization, and feature selection are applied to ensure data quality and consistency before analysis.

Model Training: A machine learning model is initially trained using historical game interaction data generated through the simulation environment. This training process enables the system to learn patterns related to learner behavior and performance.

Predictive Analysis (Real-Time): Once trained, the model is used to analyze new

incoming data in real time. It predicts potential learning issues such as low engagement, performance decline, or difficulty in task completion.

Parental Dashboard Visualization: The prediction results are presented through a parental dashboard. The dashboard provides simple and clear visual insights into learner progress and engagement, allowing parents to easily understand their child's learning behavior.

2.1.1. Technologies, Techniques, and Algorithms

The system is implemented using a combination of modern technologies, data-driven techniques, and machine learning algorithms.

Table 4: Technologies, Techniques, and Algorithms

Category	Component	Description
Technologies	Flutter	Used for cross-platform mobile application development
	Android Studio	Development and testing environment for the mobile app
	Firebase	Backend service for real-time database, storage, and authentication
	PyCharm	Used for developing and training machine learning models
Techniques	Data Collection	Collects game interaction data (time, attempts, difficulty, score)
	Data Preprocessing	Cleaning, normalization, and feature selection
	Predictive Analytics	Identifies patterns and predicts learner performance
	Real-Time Monitoring	Continuously tracks learner activity
	Parental Feedback	Provides insights and alerts to parents
Algorithms	Logistic Regression	Used for binary classification (intervention prediction)
	Artificial Neural Network (ANN)	Used to capture complex behavioral patterns

Justification of Technology Choices

Flutter was selected for frontend mobile development due to its high performance on low-end Android devices, which is essential considering the target user base in Sri Lanka where mid-range and budget devices are commonly used. Additionally, Flutter's single codebase reduces development time and maintenance effort.

Firebase was chosen as the backend platform because it provides real-time database synchronization, which is critical for continuous learner monitoring. Its managed infrastructure eliminates the need for server maintenance, making it suitable for an academic project with limited resources.

Logistic Regression was selected as the primary classification algorithm due to its

simplicity and interpretability, allowing clear understanding of how input features influence prediction outcomes. Artificial Neural Networks (ANN) were incorporated as a complementary model to capture complex, non-linear relationships in learner behavior that simpler models may not detect. This combination enables a balanced comparison between model interpretability and predictive performance, as supported by existing machine learning research [3].

2.1.2. Game-Based Data Generation and Simulation Approach

The proposed system incorporates a game-based data generation module that functions as both a learning environment and a data collection mechanism. This module is designed to simulate realistic learner interactions while children engage with game-based activities within the application.

Through this simulation approach, the system captures key behavioral data such as time spent on tasks, number of attempts, difficulty levels, and achieved scores. These data are generated dynamically during gameplay, ensuring that the dataset reflects real-world learner behavior rather than relying on static or pre-existing datasets.

In addition, the simulation-based training approach enables the machine learning model to be trained on context-specific interaction patterns. This improves the relevance and accuracy of predictions, as the model learns directly from the type of data it will encounter during real system usage.

Overall, this approach enhances the reliability of the predictive model and ensures that the system remains adaptive, practical, and aligned with real-world learning scenarios.

2.1.3. Cognitive Skill Enhancement through Game Design

In addition to serving as a data collection mechanism, the developed game-based simulator is designed to enhance cognitive skills, particularly thinking speed, memory, and attention in learners.

The game consists of five progressive difficulty levels, namely Recruit, Cadet, Officer, Commander, and Admiral. Each level is structured to gradually increase cognitive challenge by reducing the available time to complete tasks and increasing the complexity of sequences. As learners progress to higher levels, they are required to respond more quickly and accurately, thereby improving their processing speed and decision-making ability.

The system dynamically adjusts gameplay parameters such as sequence length, time limits, and speed multipliers to create a progressively challenging environment. This design encourages learners to improve reaction time, pattern recognition, and problem-solving skills through repeated interaction.

Furthermore, the game records detailed performance metrics, including accuracy, number of correct sequences, total attempts, and time taken. These metrics not only support machine learning model training but also provide valuable insights into the learner's cognitive development over time.

By integrating cognitive skill enhancement with real-time data collection, the game-based simulator functions as both an engaging learning tool and an effective

mechanism for generating meaningful behavioral data for predictive analysis.

Table 5: Operational Game Variables and Their Cognitive Skill Implications

Code Feature	**Research Meaning**
timeLimit ↓	Faster thinking required
sequenceLength ↑	Memory improvement
speedMultiplier ↑	Reaction speed
accuracy	Performance measurement

2.1.4. Implementation

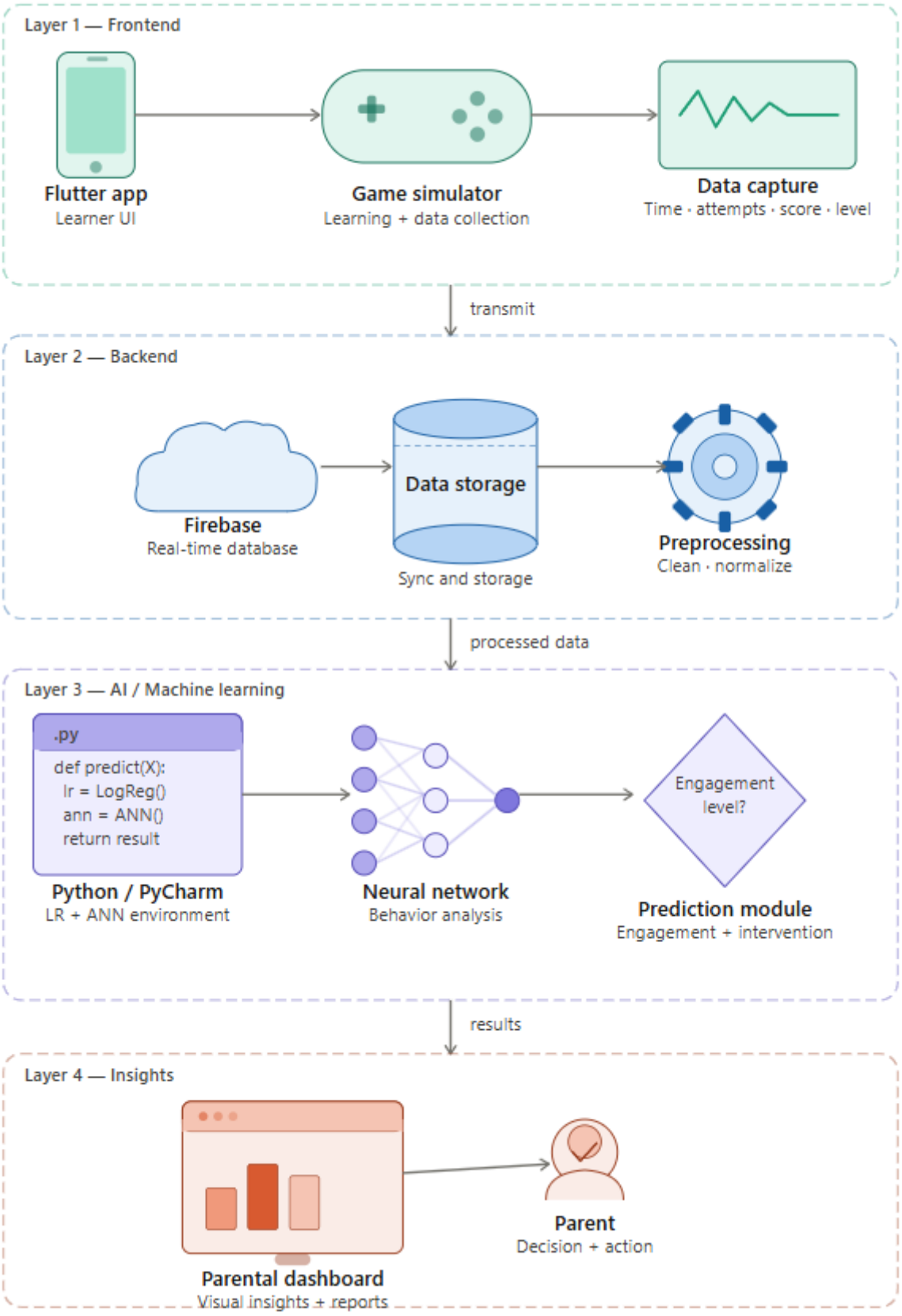

Figure 2: Implementation

The proposed system follows a modular architecture that integrates mobile application interfaces, cloud-based data management, and machine learning components to provide predictive learning insights and parental support. The architecture ensures a smooth flow of data from learner interaction to actionable insights.

2.1.5. Implementation overview of the System

The platform focuses on collecting game-based interaction data from learners, processing these data using machine learning techniques, and providing simplified insights to parents. The system is designed to support early identification of learning difficulties and enable timely intervention through a user-friendly interface.

The system incorporates a game-based simulator that serves both as a learning tool and a data collection mechanism. The simulator generates real-time interaction data, such as time spent, number of attempts, difficulty levels, and scores, which are used to train and evaluate the predictive model. This approach ensures that the system is based on realistic user behavior rather than static datasets.

2.1.6. System Architecture Components

Frontend (UI) Layer

The frontend is implemented as a mobile application using Flutter. This layer serves as the primary interface for learners and parents. Provides game-based learning activities for students, collects interaction data such as time spent, attempts, difficulty level, and scores, displays results through a simplified parental dashboard, ensures a user-friendly and accessible interface

Backend (Server) Layer

The backend is implemented using Firebase, which supports real-time data handling. Receives data from the mobile application. manages data synchronization in real time, handles communication between frontend and AI modules, ensures efficient and scalable system performance

Database Management

Firebase provides a centralized database for storing learner interaction data. Stores game interaction data securely, maintains consistency and availability of data, supports real-time updates for continuous monitoring

Predictive Analysis & AI Module

The AI module is developed using Python tools in PyCharm and is responsible for analyzing learner behavior. Performs data preprocessing (cleaning and normalization). uses machine learning algorithms such as Logistic Regression and Artificial Neural Networks, identifies patterns in learner engagement and performance, predicts potential learning issues such as low engagement or performance decline

Implementation Workflow

The overall system workflow operates as a continuous cycle:

1. Learners interact with game-based activities through the mobile app

2. Interaction data are collected and sent to the backend
3. Data are stored and preprocessed for analysis
4. The AI model generates predictions based on learner behavior
5. Results are displayed on the parental dashboard as simple insights
6. Alerts are generated for unusual patterns or performance drops
7. New data are used to improve the model over time through a feedback loop

2.1.7. Dataset description

The dataset used in this study is generated through a game-based simulation environment designed to capture realistic learner interaction data. Instead of relying on publicly available datasets, a custom dataset was created by recording user interactions during gameplay, ensuring that the data reflect real-world behavioral patterns relevant to learners with Down syndrome.

The dataset consists of several key features collected during game sessions, including time spent on tasks, number of attempts, difficulty level of the game, and achieved scores. These features are selected as they directly represent learner engagement, persistence, and performance. Additionally, a target variable is defined to indicate whether the learner requires intervention, which is used for training the predictive model.

Data collection is performed in real time as users interact with the game-based learning activities. The collected data are stored in a structured format and undergo preprocessing steps such as data cleaning, normalization, and encoding to ensure consistency and quality for machine learning analysis.

The dataset is then divided into training and testing subsets to evaluate the performance of the predictive model. Standard evaluation metrics such as accuracy, precision, recall, and F1-score are used to assess the effectiveness of the model in predicting learner engagement and identifying potential learning issues.

Overall, the use of a simulation-based dataset enables the system to be trained on realistic interaction patterns, making the predictions more relevant and applicable to real-world learning scenarios.

Table 6: Dataset Features

Feature	Description
Time Spent	Duration taken to complete a task
Number of Attempts	Number of tries to complete a level
Difficulty Level	Game difficulty (Easy, Medium, Hard)
Score	Points achieved in the game
Needs Intervention	Target variable (Yes/No)

2.1.8. AI Model Training Process

The predictive component of the proposed system is developed using a structured machine learning pipeline to ensure accurate and reliable prediction of learner engagement and performance. The training process consists of the following steps:

Step 1: Data Collection
Interaction data are collected through the game-based simulation environment while learners engage in activities. The dataset includes features such as time spent, number of attempts, difficulty level, and score, which represent learner behavior and performance.

Step 2: Data Preprocessing
The collected data are preprocessed to improve quality and consistency. This includes handling missing values, removing noise, normalizing numerical features, and encoding categorical variables such as difficulty level.

Step 3: Feature Selection
Relevant features that significantly influence learner performance are selected. These include time spent, attempts, difficulty level, and score, which are used as input variables for the model.

Step 4: Dataset Splitting
The dataset is divided into training and testing subsets (e.g., 80% training and 20% testing) to evaluate the model's performance on unseen data.

Step 5: Model Training
Supervised machine learning algorithms, including Logistic Regression and Artificial Neural Networks, are trained using the training dataset. The model learns patterns that relate learner interaction data to the target variable (e.g., need for intervention).

Step 6: Model Evaluation
The trained model is evaluated using the testing dataset. Performance metrics such as accuracy, precision, recall, and F1-score are calculated to assess prediction quality.

Step 7: Model Optimization
The model is refined by adjusting parameters and improving feature representation to enhance prediction performance and reduce errors.

Step 8: Deployment for Prediction
Once validated, the trained model is integrated into the system. During real-time usage, new interaction data are fed into the model to generate predictions on learner engagement and potential learning issues.

Step 9: Continuous Improvement
New data collected from users are periodically used to retrain and improve the model, ensuring that the system remains adaptive and accurate over time.

Overall, this step-by-step training process ensures that the predictive model is robust, reliable, and capable of supporting parental decision-making through accurate insights.

2.2.Feature Engineering

Feature engineering plays a critical role in improving the performance and accuracy of the predictive model. In this study, key behavioral features were carefully selected based on their relevance to learner engagement and cognitive performance.

The selected features include:

Time Spent on Tasks:
This feature represents the duration taken by the learner to complete a game activity. Longer completion times may indicate difficulty in understanding tasks or reduced cognitive processing speed. Conversely, shorter times may indicate better comprehension and faster thinking ability.

Number of Attempts:
The number of attempts reflects how many times a learner tries to complete a task. A higher number of attempts often indicates struggle or uncertainty, while fewer attempts suggest better confidence and understanding.

Difficulty Level:
The difficulty level represents the complexity of the game at each stage. Higher levels require greater cognitive effort, faster responses, and improved memory. This feature helps the model understand learner capability across different challenge levels.

Score:
The score reflects overall performance and accuracy during gameplay. Higher scores indicate better task completion and understanding, while lower scores suggest potential learning difficulties.

These features collectively capture both performance and behavioral aspects of the learner. By combining them, the model can identify patterns such as slow response with low score (possible difficulty) or high attempts with high difficulty (possible challenge adaptation).

Feature engineering enhances the predictive capability of the system by transforming raw interaction data into meaningful inputs. This ensures that the machine learning model can effectively learn patterns and make accurate predictions regarding learner engagement and intervention needs.

2.3. Model Comparison (Logistic Regression vs ANN)

Two machine learning models were used in this study: Logistic Regression and Artificial Neural Networks (ANN). Each model offers unique advantages and is suitable for different types of data patterns.

Logistic Regression is a simple and interpretable model that works well for binary classification problems, such as predicting whether a learner needs intervention. It provides clear insights into how each feature influences the prediction.

On the other hand, Artificial Neural Networks can capture complex and non-linear relationships in data. This makes ANN more suitable for identifying hidden behavioral patterns that cannot be easily detected by linear models.

The comparison between the two models is summarized in Table 3.

Criteria	Logistic Regression	Artificial Neural Network (ANN)
Accuracy	Moderate to High	High
Speed	Fast training and prediction	Slower due to complexity
Interpretability	High (easy to understand)	Low (black-box model)
Complexity	Low	High
Handling Non-linearity	Limited	Strong
Suitability	Simple classification	Complex behavior prediction

Figure 3: The comparison between the two models is summarized

The results indicate that Logistic Regression provides reliable baseline performance with high interpretability, making it suitable for explaining predictions to caregivers. In contrast, ANN improves prediction accuracy by capturing complex behavioral relationships, especially for diverse learner patterns.

Therefore, a combined approach using both models enhances the robustness and effectiveness of the system.

2.4. Commercialization Aspects

The proposed system demonstrates strong commercialization potential as an AI-driven educational support tool designed for learners with Down syndrome and their caregivers. By combining game-based learning, real-time data analytics, and predictive insights, the system addresses a growing need for personalized and accessible educational technologies.

From a market perspective, the product targets multiple user groups, including parents (caregivers), special education centers, therapy providers, and schools that support children with neurodevelopmental conditions. The increasing adoption of digital learning platforms and the rising awareness of inclusive education create a favorable environment for the deployment of such a solution.

The system can be commercialized as a mobile application using a Software-as-a-Service (SaaS) model. A freemium approach can be adopted, where basic game-based learning features are provided free of charge, while advanced functionalities such as detailed analytics, predictive insights, and personalized reports are offered through subscription plans. This model ensures accessibility while maintaining financial sustainability.

Additionally, the platform can be extended through partnerships with educational institutions, non-governmental organizations, and healthcare providers. Such collaborations can support wider adoption, particularly in regions with limited access

to specialized learning tools. Localization features, including language support and culturally relevant content, can further enhance market reach.

The system also supports scalability through its cloud-based backend, allowing it to accommodate a growing number of users without significant infrastructure changes. Continuous data collection and model improvement provide opportunities for future upgrades, such as adaptive learning pathways and enhanced intervention strategies.

Furthermore, ethical considerations such as data privacy and secure access (e.g., PIN-based parental control) strengthen user trust, which is a critical factor for commercialization in educational and healthcare-related technologies.

3. Testing

The proposed system was evaluated using multiple software testing techniques to ensure reliability, functionality, and predictive performance. The testing process focused on verifying individual components, interactions between modules, overall system behavior, and the accuracy of the machine learning model. The main testing approaches used include unit testing, integration testing, functional testing, and accuracy testing.

Unit Testing
Unit testing was conducted to verify the correctness of individual components of the system. Each module, such as data collection, preprocessing, prediction, and dashboard display, was tested independently to ensure that it performs as expected.

Integration Testing
Integration testing was performed to evaluate the interaction between different system modules. This includes testing the data flow between the mobile application, backend (Firebase), and the AI prediction module to ensure seamless communication.

Functional Testing
Functional testing was carried out to validate the overall system behavior against the specified requirements. This includes verifying gameplay data collection, prediction generation, dashboard visualization, and alert notifications.

Accuracy Testing
Accuracy testing was performed to evaluate the performance of the machine learning model used for prediction. The model was tested using a dataset of game interaction data, including time spent, number of attempts, difficulty level, and score.

The performance of the model was measured using standard evaluation metrics such as accuracy, precision, recall, and F1-score. These metrics help determine how effectively the model predicts learner engagement and identifies potential learning issues. The results indicate that the model provides reliable predictions suitable for supporting parental decision-making.

3.1. Test cases

Table 7: test cases

Test Case ID	Test Scenario	Input	Expected Output	Result
TC01	Game data collection	User plays game	Time, score, attempts recorded	Pass
TC02	Data storage in database	Game data submitted	Data stored in Firebase	Pass
TC03	Data preprocessing	Raw data input	Cleaned & normalized data	Pass
TC04	Model prediction	Processed data	Engagement prediction generated	Pass
TC05	Dashboard display	Prediction results	Data shown clearly to parent	Pass
TC06	Alert generation	Low score / high attempts	Alert notification triggered	Pass
TC07	Real-time update	Continuous gameplay	Data updated instantly	Pass
TC08	Integration (App → Firebase)	Data sent from app	Successfully stored in backend	Pass
TC09	Integration (Backend → AI model)	Stored data	Data passed correctly to model	Pass
TC10	UI usability	Parent uses dashboard	Easy navigation and readability	Pass

3.2. Use Case Diagram

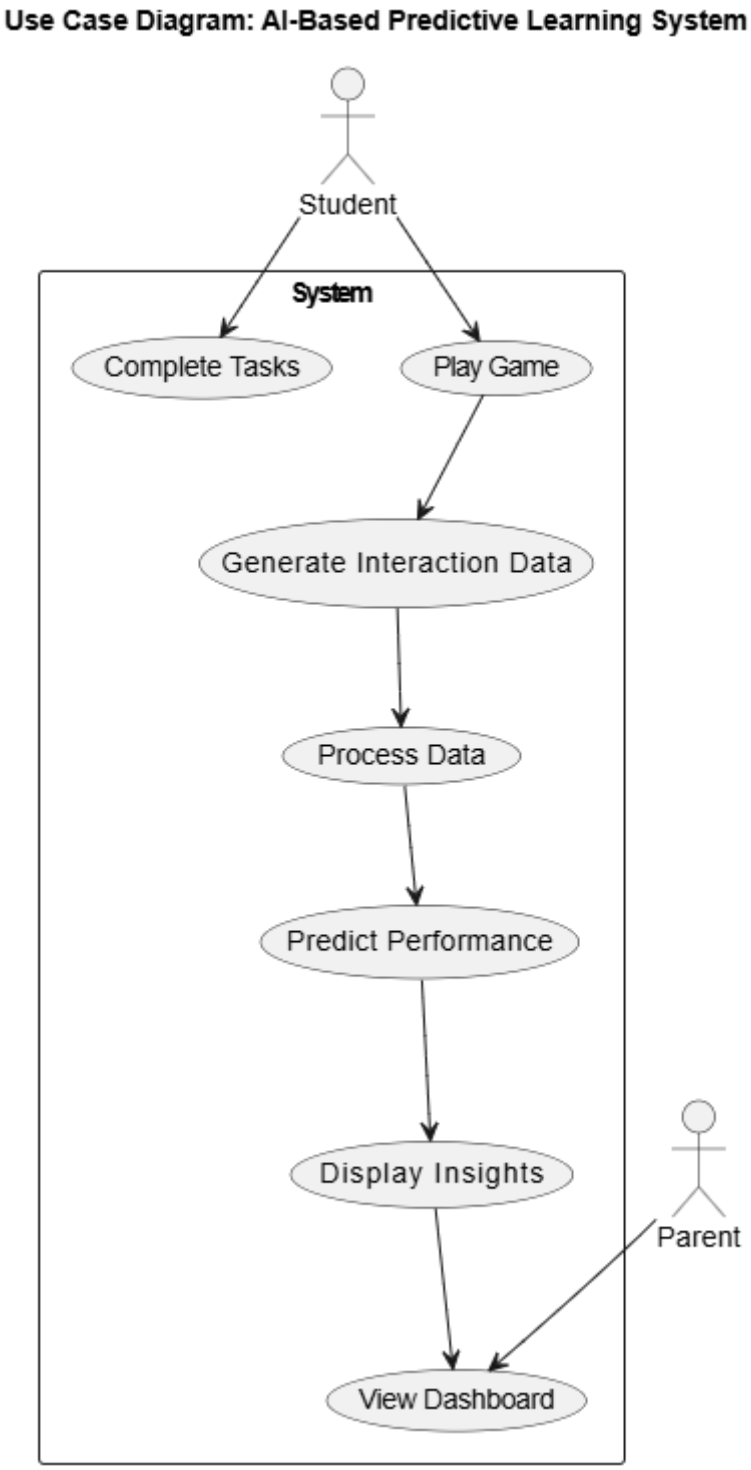

Figure 4: use case diagram

The use case diagram represents the interaction between the main users of the system and its functionalities. The system consists of two primary actors: the student and the parent.

The student interacts with the system by playing game-based learning activities and completing tasks. During these interactions, the system automatically generates and collects game interaction data.

The parent interacts with the system through a dashboard interface, where they can view learner progress, receive alerts, and access reports. These features provide simplified insights into the learner's engagement and performance.

Internally, the system processes the collected data, performs predictive analysis using machine learning models, and generates insights. These insights are then displayed to the parent and used to trigger alerts when necessary.

Overall, the use case diagram illustrates how the system supports learner interaction while providing meaningful feedback and decision-making support to parents.

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

The functional requirements define the core features and operations that the system must perform to meet user needs. The proposed system includes the following functionalities:

- User Interaction through Game-Based Learning: The system shall allow students to engage in game-based learning activities designed to capture interaction data such as time spent, number of attempts, difficulty level, and scores.
- Real-Time Data Collection: The system shall continuously collect and record learner interaction data during gameplay without interrupting the user experience.
- Data Storage and Management: The system shall store collected data securely in a centralized database and ensure data consistency and availability for processing.
- Data Preprocessing: The system shall process raw data using techniques such as cleaning, normalization, and feature selection before analysis.
- Predictive Analysis: The system shall apply machine learning algorithms to analyze learner behavior and predict engagement levels and potential learning issues.
- Parental Dashboard: The system shall provide a dashboard for parents to view learner progress, engagement levels, and performance insights in a clear and simplified manner.

4.2. Non-Functional Requirements

The non-functional requirements define the quality attributes and constraints that ensure the system operates effectively and efficiently.

- Usability: The system should provide a simple, intuitive, and user-friendly interface suitable for parents and children with minimal technical knowledge.
- Performance: The system should process data and generate predictions with minimal delays to support near real-time feedback.
- Scalability: The system shall support an increasing number of users and data without significant performance degradation.
- Reliability: The system shall operate consistently without failures and ensure accurate data processing and prediction.
- Availability: The system shall always be accessible with minimal downtime.
- Maintainability: The system shall be designed in a modular manner to allow easy updates, debugging, and improvements.
- Compatibility: The system should function across different Android devices and screen sizes.
- Privacy: The system shall ensure that learner data are handled ethically and used only for intended purposes, maintaining parental trust. Access to sensitive information, such as performance insights and predictive results,

shall be restricted through a secure PIN-based authentication mechanism for parents. This ensures that only authorized users can view detailed learner data, while preventing unintended access by students. The system is designed to present information in a controlled manner to protect the learner's privacy and emotional well-being.

Measurable Acceptance Criteria for Key Non-Functional Requirements

Non-Functional Requirement	Measurable Acceptance Criteria
Performance	Prediction results shall be generated and displayed on the parental dashboard within 3 seconds of new interaction data being submitted.
Availability	The system shall maintain at least 99% uptime during the pilot testing period.
Scalability	The system shall support a minimum of 100 concurrent user sessions without degradation in response time.
Usability	A System Usability Scale (SUS) score of 70 or above shall be achieved during user acceptance testing with parents and caregivers.
Reliability	The predictive model shall achieve a minimum accuracy of 80% and an F1-score of 0.75 or above on the test dataset.
Security	No unauthorized data access events shall occur during penetration testing.

Figure 5: Measurable Acceptance Criteria for Key Non-Functional Requirements

5. SOCIAL, ETHICAL, AND SECURITY CONSIDERATIONS

5.1. Social Impact

This system is designed with a strong social purpose: to reduce educational inequity for children with Down syndrome in Sri Lanka, a group that has historically been underserved by mainstream educational technology. By empowering parents with real-time, data-driven insights, the platform promotes family-centered care, which is recognized as a key factor in improving outcomes for children with intellectual disabilities [20], [22]. The system also supports teachers and therapists in making informed, evidence-based decisions, reducing the reliance on subjective observations and enabling a more structured and equitable learning environment.

A key social limitation of the system is the digital divide: families without access to smartphones, reliable internet, or sufficient digital literacy may not be able to benefit fully from the platform. To address this, the system is designed to function on low-end Android devices and under low-bandwidth conditions. Future work should include community-based training programs for parents and educators to support adoption in resource-limited settings.

5.2. Ethical Considerations

5.2.1. Informed Consent

All data collection involving children with Down syndrome requires explicit informed consent from parents or legal guardians prior to system use. Consent forms will clearly explain what data are collected, how they are used, and the rights of participants to withdraw at any time. This process will adhere to ethical standards recommended by the Sri Lanka Medical Association and relevant institutional review boards [17].

5.2.1. Algorithmic Bias.

A critical ethical concern in AI-based educational systems is the risk of algorithmic bias. Since the training dataset is simulation-generated, it may not fully capture the behavioral diversity among children with Down syndrome, who exhibit a wide range of cognitive profiles. Biased predictions could result in incorrect or stigmatizing intervention alerts, potentially affecting a child’s educational experience negatively.

To mitigate this, the model will be validated against diverse real-world interaction data during pilot testing, and fairness metrics will be evaluated alongside accuracy metrics [3].

5.2.3. Data Minimization and Purpose Limitation

In line with ethical AI principles, the system will collect only the minimum data necessary for its predictive function. Learner data will not be shared with third parties, used for advertising, or repurposed beyond the stated objectives of learning support and parental feedback. All data use will be transparently communicated to users through an accessible privacy policy.

5.2.4. Learner Dignity and Well-being

The system avoids presenting performance data in ways that could be stigmatizing or distressing to the child. Dashboard language is designed to be constructive and supportive rather than deficit-focused. Labels such as “needs intervention” are displayed only to parents and caregivers, never to the learner, and are framed as opportunities for additional support rather than indicators of failure.

5.3. Security Considerations

5.3.1. Authentication and Access Control

The system implements a role-based access control (RBAC) model. Parents access the dashboard via a secure PIN-based authentication mechanism, while teacher and therapist accounts are managed through Firebase Authentication with email verification. Administrative access is restricted to authorized system administrators only. This ensures that sensitive learner data are accessible only to those with a legitimate need.

5.3.2. Data Encryption and Storage Security

All data transmitted between the mobile application and the Firebase backend are encrypted using TLS (Transport Layer Security) 1.2 or higher. Data at rest in Firebase are encrypted using AES-256 encryption. Firebase Security Rules are configured to enforce that each user can only read and write their own data, preventing unauthorized cross-user data access.

6. System Limitations and Future Work

6.1. System Limitations

The proposed system demonstrates the potential of integrating game-based learning with AI-driven predictive analytics for supporting learners; however, several limitations should be acknowledged.

Firstly, the system relies primarily on a simulation-generated training dataset. Although the game-based simulator was designed to produce realistic interaction data, it may not fully capture the behavioral diversity of real-world learners, particularly learners with different cognitive profiles such as Down syndrome. As a result, the predictive model's performance may vary when applied to real-world populations with characteristics that differ from the training data.

Secondly, the current implementation is based on a single game module. While this module effectively captures essential behavioral indicators such as time spent, number of attempts, difficulty level, and score, it does not represent the full spectrum of learning activities. Consequently, the model may benefit from additional diversity in task types to improve robustness and generalizability.

Another limitation is the absence of real-world deployment and large-scale user testing. The system has been evaluated primarily through controlled experiments and simulation-based scenarios. Therefore, its usability, scalability, and long-term effectiveness in real educational or therapeutic environments remain to be validated.

In addition, the system currently operates only on Android devices, which limits accessibility for users on other platforms such as iOS. Furthermore, certain functionalities such as real-time monitoring and dashboard visualization are dependent on a stable internet connection, which may be a constraint in rural or low-resource environments.

Finally, the predictive models used in this study (Logistic Regression and Artificial Neural Networks) are relatively basic compared to more advanced deep learning or ensemble approaches. While they provide interpretability and acceptable performance, they may not fully capture complex and non-linear behavioral patterns.

6.2. Future Work

To address these limitations, several future enhancements are proposed.

A key direction is the integration of real-world datasets through collaboration with schools, therapy centers, and special education institutions. This would significantly improve model generalization and ensure better adaptation to real learner behavior.

The system can also be expanded to include multiple game modules covering different cognitive and educational tasks. This would enrich behavioral data diversity and improve predictive accuracy while enhancing learner engagement.

Future work may also involve the adoption of advanced machine learning techniques, such as deep learning and ensemble models, to better capture complex behavioral relationships and improve prediction performance.

In addition, integrating multimodal data sources such as speech recognition, facial expression analysis, and possibly physiological signals would enable a more comprehensive understanding of learner engagement and emotional state.

From an accessibility perspective, extending the system to support cross-platform deployment (including iOS and web-based interfaces) would significantly increase usability. Offline or low-bandwidth functionality could also be introduced to support rural and resource-limited environments.

Furthermore, the caregiver dashboard can be enhanced with intelligent recommendation systems, providing actionable insights and intervention strategies based on predictive analytics.

Finally, large-scale longitudinal studies should be conducted to evaluate the long-term educational impact of the system, including improvements in engagement, learning outcomes, and caregiver satisfaction.

7. Results and Discussion

7.1. Experimental Setup

The evaluation of the proposed system was conducted using a simulation-based dataset generated through the game-based learning environment. The dataset consists of learner interaction records captured during gameplay, including key behavioral and performance features such as time spent, number of attempts, difficulty level, score, and accuracy.

The dataset was divided into training and testing subsets using an 80:20 ratio to ensure unbiased evaluation. Two machine learning models, Logistic Regression and Artificial Neural Networks (ANN), were trained and tested using the same dataset to enable a fair comparison of performance. The evaluation focused on the system's ability to predict learner engagement levels, identify potential learning difficulties, and support early intervention.

7.2. Model Performance Analysis

The results demonstrate that both machine learning models can classify learner behavior into meaningful categories such as needs intervention "yes" or "no". However, their performance differs based on their underlying computational characteristics.

The Logistic Regression model achieved stable and consistent performance, providing reliable baseline predictions. Its linear nature allows for clear interpretation of how each feature contributes to the final prediction. This makes it particularly suitable for explaining results to non-technical users such as parents and caregivers.

In contrast, the Artificial Neural Network (ANN) model achieved higher prediction accuracy by capturing complex and non-linear relationships between behavioral features. The ANN model demonstrated superior performance in identifying subtle interaction patterns that are not easily captured by linear models.

7.3. Evaluation Metrics

The performance of the models was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. Accuracy measures the overall correctness of predictions, while precision and recall provide deeper insights into the model's ability to correctly identify learners who require intervention.

The ANN model achieved higher recall and F1-score values compared to Logistic Regression, indicating its effectiveness in detecting learners at risk. This is particularly important in educational applications, where failing to identify a struggling learner can negatively impact learning outcomes.

The Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC) further confirm the performance difference between the models. The ANN model achieved an AUC value of 0.97, compared to 0.91 for Logistic Regression, demonstrating its superior ability to distinguish between different learner states.

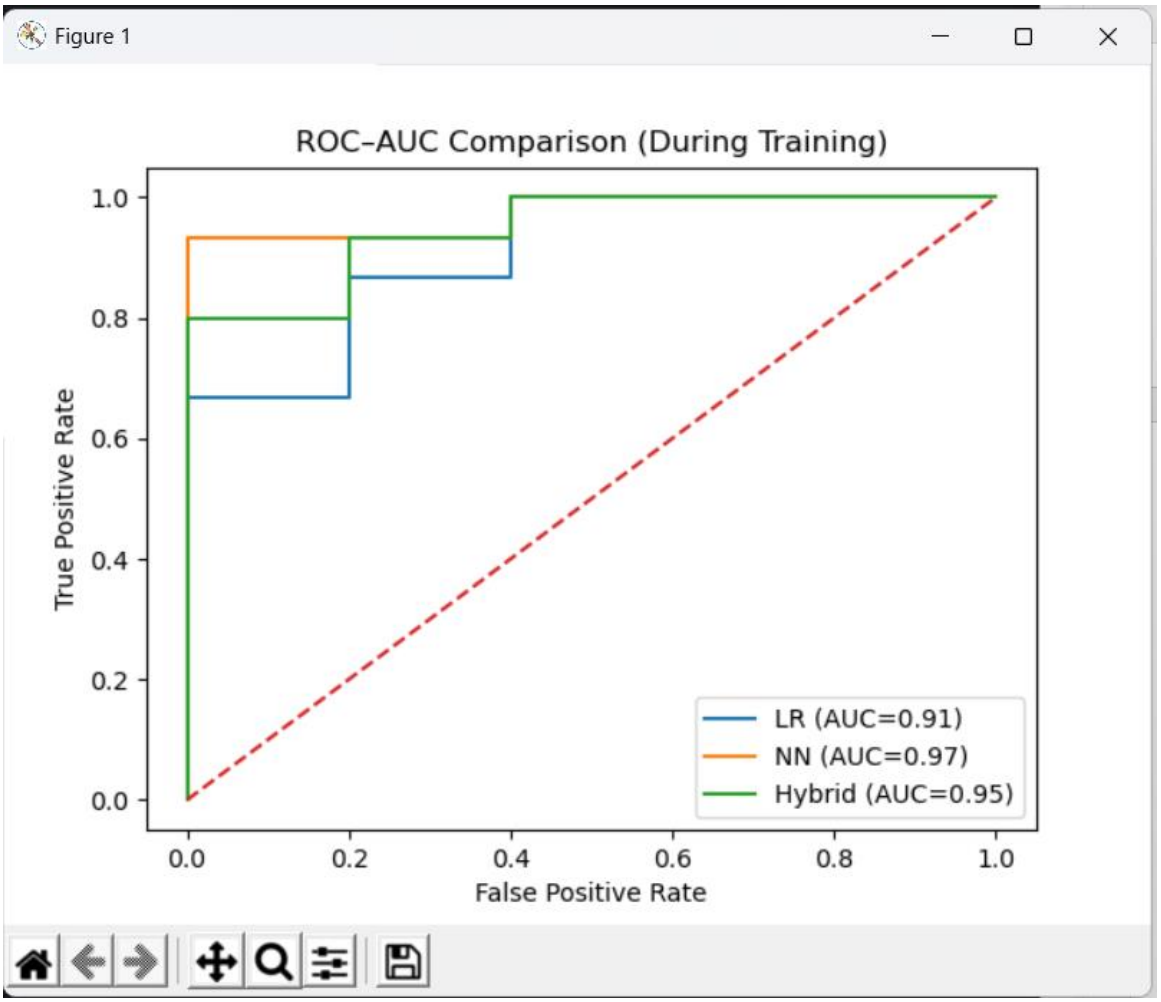

Figure 6: ROC-AUC Comparison (During Training)

X-axis (False Positive Rate) → How often the model incorrectly predicts positive
Y-axis (True Positive Rate) → How often the model correctly predicts positive
Each curve = one model

- Blue → Logistic Regression (AUC = 0.91)
- Orange → Neural Network (AUC = 0.97)
- Green → Hybrid Model (AUC = 0.95)

7.3. Behavioral Pattern Analysis

One of the key contributions of this research is the identification of meaningful behavioral patterns from game interaction data.

7.3.1 High-Risk Behavior Patterns

The system identified several patterns associated with learners who require intervention:

- High number of attempts combined with low scores
- Increased time spent with minimal performance improvement
- Repeated failures at higher difficulty levels

These patterns indicate cognitive difficulty, reduced engagement, or lack of understanding, which are critical indicators for early intervention.

7.3.2 Positive Learning Patterns

The system also identified patterns associated with effective learning:

- High scores with low number of attempts
- Consistent performance improvement across levels
- Reduced time spent with increased accuracy

These patterns suggest strong comprehension, efficient learning, and increased confidence.

7.3.3 Intermediate Patterns

Some learners exhibited mixed behavior, such as:

- High time spent but high scores
- Moderate attempts with moderate performance

These cases indicate learners who are progressing but may require occasional support.

7.4. Effectiveness of Game-Based Data Collection

The game-based simulator proved to be highly effective in generating realistic and meaningful interaction data.

Advantages:

- Captures real-time behavioral data
- Provides continuous data streams
- Reflects natural learning behavior

Unlike traditional datasets, which are often static, the simulation-based approach ensures that the model learns from dynamic and context-specific interactions.

7.5. Impact on Model Performance

The quality of the generated dataset directly influenced prediction accuracy. Since the data reflect actual user interactions, the model is better aligned with real-world learning scenarios.

However, it is important to note that simulation-based data may still lack certain real-

world complexities, such as emotional and environmental factors.

7.5. Discussion on Model Selection

7.5.1 Logistic Regression

Table 8: Logistic Regression advantages and limitations

Advantages	Limitations
High interpretability	Cannot capture complex patterns
Fast computation	Assumes linear relationships
Suitable for simple classification	

7.5.2 Artificial Neural Networks (ANN)

Table 9: ANN advantages and limitations

Advantages	Limitations
Captures non-linear relationships	Low interpretability
Higher prediction accuracy	Requires more computational resources
Better generalization	

7.5.3 Combined Approach

Using both models provides a balanced solution:

✅Logistic Regression for explanation

✅ANN for accuracy

This combination improves both usability and performance.

7.6. Dashboard Effectiveness and User Impact

The parental dashboard plays a crucial role in translating technical predictions into understandable insights.

Parents can easily interpret progress,

Visual summaries improve decision-making

The system successfully bridges the gap between complex AI outputs and non-

technical users.

7.8. Limitations of Results

Despite the promising results, several limitations should be acknowledged. The model was trained using simulation-generated data from a single game environment, which may limit the diversity of behavioral patterns captured. This can affect the model's ability to generalize across different learning contexts.

Additionally, the evaluation was conducted in a controlled environment without large-scale real-world testing. As a result, the long-term effectiveness and usability of the system in real educational settings remain to be validated.

7.9. Implications for Inclusive Education

The findings demonstrate that AI-driven systems can significantly improve inclusive education by:

- Supporting early detection of learning difficulties,
- Enhancing parental involvement,
- Providing personalized learning insights, this is particularly valuable for learners with Down syndrome, who require continuous monitoring and adaptive support.

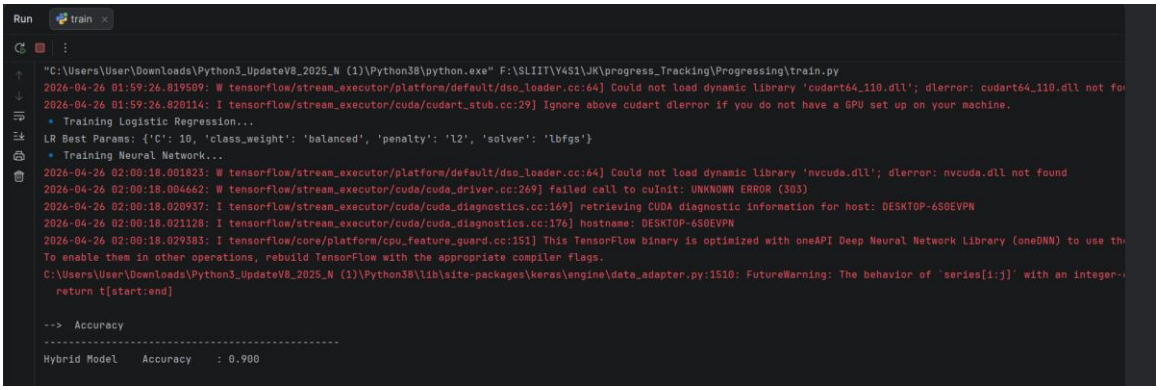

Figure 7: Model training accuracy output

7. Conclusion

This research presented the design and development of an AI-driven, game-based learning and predictive analytics system aimed at supporting learners with Down syndrome through a caregiver-centered approach. The system successfully integrates real-time data collection, machine learning techniques, and user-friendly visualization to provide meaningful insights into learner engagement and performance.

A key contribution of this study is the development of a game-based simulator that functions both as an educational tool and a data generation mechanism. By capturing real-time interaction data such as time spent, number of attempts, difficulty levels, and scores, the system enables the creation of a realistic and context-specific dataset for training predictive models. This approach addresses the common limitation of lacking domain-specific datasets in inclusive education research.

The implementation of machine learning models, including Logistic Regression and Artificial Neural Networks, demonstrated the feasibility of predicting learner behavior and identifying potential learning difficulties. The results indicate that the system can effectively support early intervention by detecting patterns associated with low engagement and performance decline.

In addition, the system emphasizes a parental (caregiver) perspective by providing a simplified dashboard and real-time alerts, enabling non-technical users to understand and respond to learner needs. The inclusion of secure access control mechanisms further ensures data privacy and ethical usage, which are critical factors in educational technologies.

Furthermore, the game design contributes to cognitive skill enhancement by improving thinking speed, memory, and attention through progressive difficulty levels. This highlights the dual benefit of the system as both an analytical and developmental tool.

8. GANTT CHART

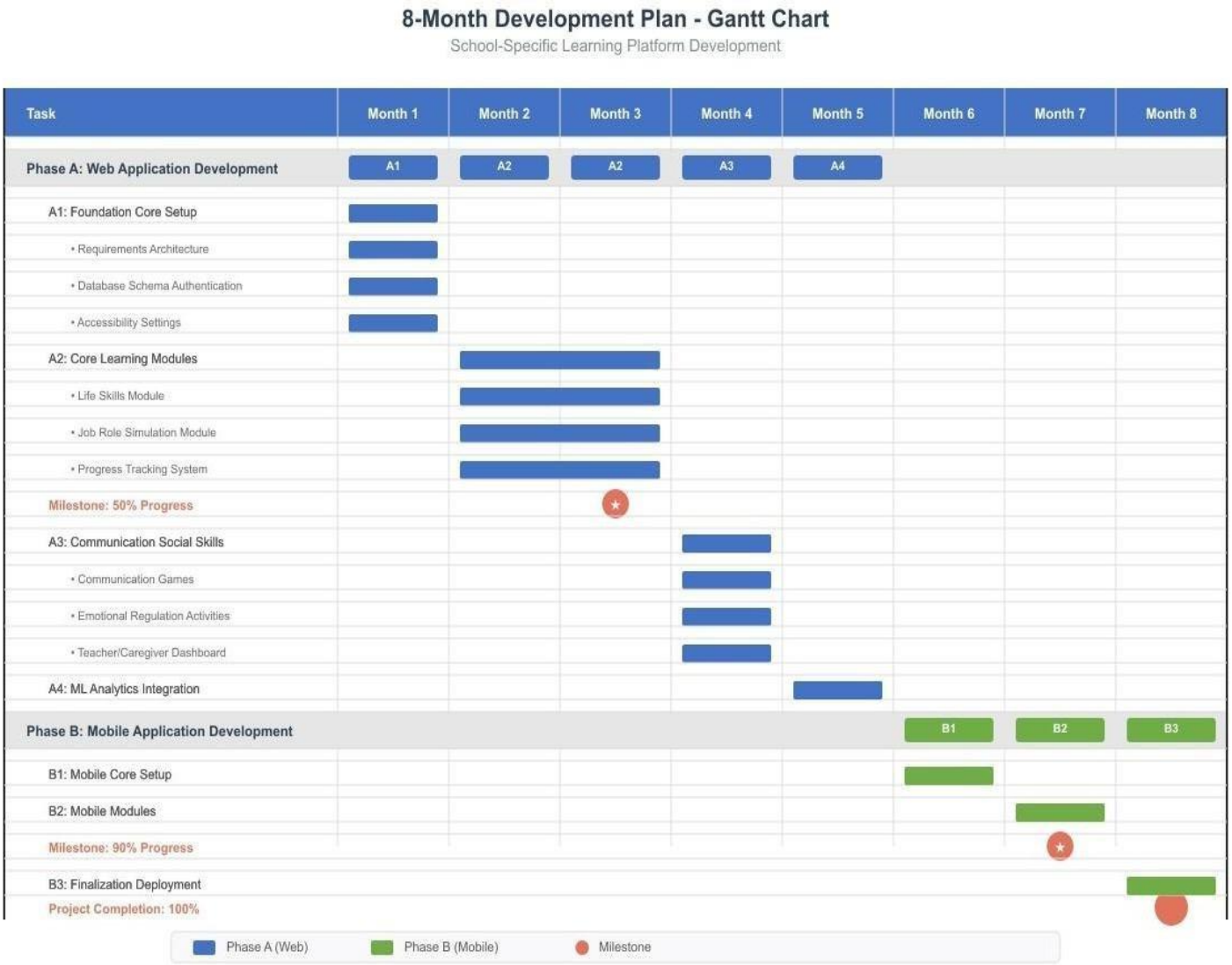

Figure 8: Gantt Chart

9. WORK BREAKDOWN STRUCTURE

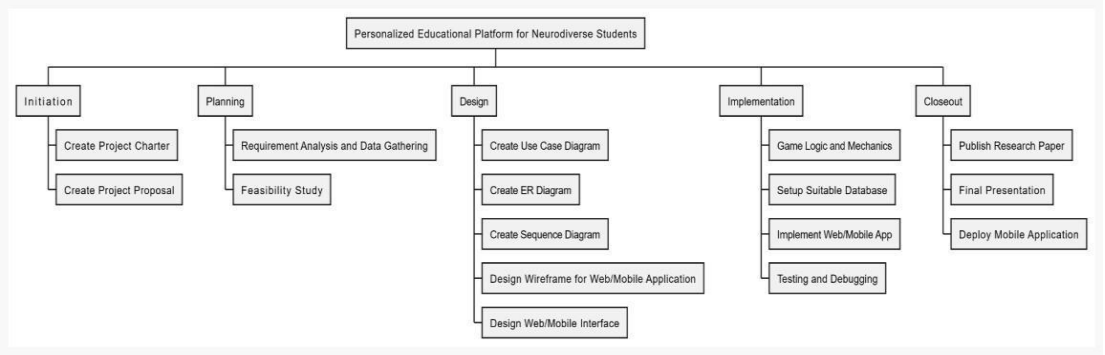

Figure 9: Work breakdown structure

10. BUDGET

Table 10: Budget

Requirements	Price (LKR)
Filed visits for data collection	**6000.00**
Internet and Wi-Fi charges	**6000.00**
Printing and Publishing	**4000.00**
Software/ML tool licenses (Python libraries)	**8000.00**
Other charges	**6000.00**
Total estimated cost	**30 000.00**

11. Commercialization

The proposed progress tracking and parental feedback system holds strong commercialization potential as it directly addresses the growing demand for personalized and inclusive educational technologies, particularly for neurodiverse and special-needs learners. By integrating AI-driven behavioral analytics and real-time feedback, the solution offers schools, therapy centers, and ed-tech companies a competitive edge in enhancing student engagement and well-being. The proactive intervention model not only improves learning outcomes but also appeals to parents and educators seeking evidence-based, data-driven tools. This creates opportunities for subscription-based licensing, partnerships with educational institutions, and scaling into broader markets such as e-learning platforms and special education services.

Reference list

[1] J. S. Brown and R. R. Burton, “Diagnostic models for procedural bugs in basic mathematics skills,” *Cognitive Science*, vol. 2, no. 2, pp. 155–192, 1978.
[2] P. H. Kahn et al., “Designing for engagement: Using adaptive interfaces for children with Down Syndrome,” *Int. J. Child-Comput. Interact.*, vol. 17, pp. 1–12, 2018.
[3] B. Kitchenham, “Predictive models in educational analytics: A systematic review,” *Educ. Inf. Technol.*, vol. 25, no. 4, pp. 3299–3324, 2020.
[4] R. E. Mayer, *Multimedia Learning*, 3rd ed. Cambridge, U.K.: Cambridge Univ. Press, 2020.
[5] “ClassDojo,” ClassDojo Inc., 2022. [Online]. Available: https://www.classdojo.com
[6] D. Kaur and A. Arora, “Artificial intelligence in education: Applications and challenges,” Int. J. Eng. Adv. Technol., vol. 9, no. 3, pp. 570–574, 2020.S. Deterding et al., “Gamification: Using game design elements in non-gaming contexts,” in *Proc. CHI 2011*, pp. 2425–2428.
[7] G. Siemens and P. Long, “Penetrating the fog: Analytics in learning and education,” *EDUCAUSE Review*, vol. 46, no. 5, pp. 30–40, 2011.

[8] S. Muncey, "How AI helps educators track student progress and outcomes," *SchoolAI*, May 9, 2025. [Online]. Available: https://schoolai.com/blog/ai-helps-educators-track-student-progress-outcomes

[9] "Predictive Analytics: Enhancing Education Through AI," *21K School*, 2025. [Online]. Available: https://www.21kschool.com/us/blog/predictive-analytics-in-education/

[10] "Predictive Analytics in Education: Boosting Student Success and Retention," *eSelf AI*, 2025. [Online]. Available: https://www.eself.ai/blog/predictive-analytics-in-education/

[11] "How AI helps educators track student progress and outcomes," *SchoolAI*, May 9, 2025. [Online]. Available: https://schoolai.com/blog/ai-helps-educators-track-student-progress-outcomes

[12] "AI in parent-teacher communication," *Artificial Intelligence +*, 2024. [Online]. Available: https://aiplusinfo.medium.com/ai-in-parent-teacher-communication-694c6d10aa8a

[13] "How Generative AI Improves Parent Involvement," *Panorama Education*, 2024. [Online]. Available: https://www.panoramaed.com/blog/how-generative-ai-improves-parent-involvement

[14] "AI Early Warning Dashboards Spot At-Risk Students," *SOLVED Consulting*, 2025. [Online]. Available: https://www.solvedconsulting.com/blog/ai-early-warning-dashboards-at-risk-students

[15] "Dashboard - SmartSocial.com," *SmartSocial*, 2025. [Online]. Available: https://smartsocial.com/dashboard/parents

[16] "AI in K-12 schools: Reports show nearly 70% of parents oppose sharing student data with artificial intelligence," *Times of India*, 2025. Available: https://timesofindia.indiatimes.com/education/news/ai-in-k-12-schools-reports-show-nearly-70-of-parents-oppose-sharing-student-data-with-artificial-intelligence/articleshow/123530598.cms.

[17] A. S. Al-Azawei, M. Serenelli, and A. Lundqvist, "Universal design for learning (UDL): A content analysis of peer-reviewed journal papers from 2012 to 2015," *J. Postsecondary Educ. Disabil.*, vol. 29, no. 4, pp. 317–332, 2016. [Online]. Available: https://doi.org/10.14434/josotl.v16i3.19295

[18] D. Kaur and A. Arora, "Artificial intelligence in education: Applications and challenges," *Int. J. Eng. Adv. Technol.*, vol. 9, no. 3, pp. 570–574, 2020.

[19] J. Lindsay, T. Dockrell, and L. Strand, "Technology to support learning for students with neurodevelopmental disorders," *Front. Educ.*, vol. 6, no. 59, pp. 1–12, 2021.

[20] M. M. Alsobhi, N. M. Khan, and H. Rahanu, "Personalized learning materials based on dyslexia types: Ontological approach," *Procedia Comput. Sci.*, vol. 14, pp. 351–362, 2012.

[21] UNESCO, "Assistive technology to support inclusive education," *UNESCO Policy Brief*, 2021. [Online]. Available: https://unesdoc.unesco.org

[22] UNESCO, "Assistive technology to support inclusive education," UNESCO Policy Brief, 2021. [Online]. Available: https://unesdoc.unesco.org